

OSDS Nearshore Impact Model

Madeline Berger

2025-03-29

Modelling OSDS Impact to Nearshore

This markdown walks through the steps of estimating the relative impact of OSDS on the nearshore of the Main Hawaiian Islands.

Input data needed:

- Point locations of all functional OSDS and the year they were permitted. We are using year permitted as a proxy for year built although it may be that some are not built right away
- Ocean mask
- Euclidean distance mask

Step 0: Clean and explore OSDS Data

Oahu Dataset

```
# check valid geometries

oahu_2023 <- oahu_2023 %>%
  mutate(valid = st_is_valid(.),
         empty = st_is_empty(.))

oahu_empty <- oahu_2023 %>%
  filter(empty == TRUE) # note these are empty to Julia

oahu_2023_c1 <- oahu_2023 %>%
  filter(empty == FALSE)

# look at distributions

hist(oahu_2023_c1$Effluent)
hist(oahu_2023_c1$Nitrogen_Flux[!is.na(oahu_2023$Nitrogen_Flux)])

max(oahu_2023_c1$Effluent, na.rm = T)

nas_effluent <- oahu_2023_c1 %>%
  filter(is.na(Effluent)) # 5 once we take out the non valid geometries
```

```

nas_N <- oahu_2023_cl %>%
  filter(is.na(Nitrogen_Flux)) # 5 more, same ones as above

# check on stats
max(oahu_2023_cl$Effluent, na.rm = T)
min(oahu_2023_cl$Effluent, na.rm = T) # this is zero
mean(oahu_2023_cl$Effluent, na.rm = T) #533
median(oahu_2023$Effluent, na.rm = T)

zeros <- oahu_2023_cl %>%
  filter(Effluent == 0 ) # removed another 10 observations, just to note

oahu_nas <- oahu_2023_cl %>%
  filter(is.na(Effluent)) # 5

mapview(zeros, col.regions = "blue")+
  mapview(oahu_nas, col.regions = "green")

injection_wells <- oahu_2023_cl %>%
  filter(PermitType == "Injection Well")

mapview(injection_wells)

layout(mat = matrix(c(1,2),2,1, byrow=TRUE), height = c(1,8))
par(mar=c(0, 3.1, 1.1, 2.1))
boxplot(log(oahu_2023$Effluent) , horizontal=TRUE , ylim=c(4,10), xaxt="n" , col=rgb(0.8,0.8,0,0.5) , f
par(mar=c(4, 3.1, 1.1, 2.1))
hist(log(oahu_2023$Effluent) , breaks=40 , col=rgb(0.2,0.8,0.5,0.5) , border=F , main="" , xlab="Log of

#mapview(nozeros, zcol = "Effluent")

# ok so effluent has a crazy range. Need to:
# remove injection wells and anything after 2019
# multiply yearly effluent by the years in service, which = 2019 - year
# multiply N by effluent

osds_2015_2019 <- oahu_2023_cl %>%
  filter(Year < 2020) %>%
  filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2015_2019 = (2020 - Year) - (2015 - Year)) %>%
  mutate(total_N_toenv2015_2019 = years_active_2015_2019 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2025_2019 = years_active_2015_2019 * 365 * Effluent) %>%
  mutate(Nflux_kgpd_5yr_avg = total_N_toenv2015_2019 / (365 *5)) %>%
  mutate(Effluent_galpd_5yr_avg = total_Eff_toenv2025_2019 / (365*5))

# recheck stats after year and type filters

hist(osds_2015_2019$Effluent, breaks = 30)

```

```

hist(log(osds_2015_2019$Effluent)) # can see distribution better this way
# nicer code
# Layout to split the screen
layout(mat = matrix(c(1,2),2,1, byrow=TRUE), height = c(1,8))
par(mar=c(0, 3.1, 1.1, 2.1))
boxplot(log(osds_2015_2019$Effluent) , horizontal=TRUE , ylim=c(4,10), xaxt="n" , col=rgb(0.8,0.8,0,0.5))
par(mar=c(4, 3.1, 1.1, 2.1))
hist(log(osds_2015_2019$Effluent) , breaks=40 , col=rgb(0.2,0.8,0.5,0.5) , border=F , main="" , xlab="L

max(osds_2015_2019$Effluent, na.rm = T) #14150
min(osds_2015_2019$Effluent, na.rm = T) #39
mean(osds_2015_2019$Effluent, na.rm = T) # 582
median(osds_2015_2019$Effluent, na.rm = T) #400

# questions - does Effluent scale with N or P flux? Plot, and see read me for documentation

plot(osds_2015_2019$Effluent, osds_2015_2019$Nitrogen_Flux)
plot(osds_2015_2019$Effluent, osds_2015_2019$Phosphorous_Flux)

# create versions of the points for other scenarios: with injection wells, and for previous time period

osds_2015_2019_IW <- oahu_2023_c1 %>%
  filter(Year < 2020 | is.na(Year)) %>% # this grabs the iws
  mutate(Year = ifelse(is.na(Year),2009,Year)) %>% # so we can count iws
  #filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2015_2019 = (2020 - Year) - (2015 - Year)) %>%
  mutate(total_N_toenv2015_2019 = years_active_2015_2019 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2025_2019 = years_active_2015_2019 * 365 * Effluent) %>%
  mutate(Nflux_kgpd_5yr_avg = total_N_toenv2015_2019 / (365 *5)) %>%
  mutate(Effluent_galpd_5yr_avg = total_Eff_toenv2025_2019 / (365*5)) %>%
  st_transform(., crs = crs(ocean_mask_oahu))

max(osds_2015_2019_IW$Effluent, na.rm = T) #539941.2
min(osds_2015_2019_IW$Effluent, na.rm = T) #39
mean(osds_2015_2019_IW$Effluent, na.rm = T) # 1681
median(osds_2015_2019_IW$Effluent, na.rm = T) #420

nastest <- osds_2015_2019_IW[is.na(osds_2015_2019_IW$Year),]# all nas for year are injection wells so t

osds_2010_2014 <- oahu_2023_c1 %>%
  filter(Year < 2015) %>%
  filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2010_2014 = (2015 - Year) - (2010 - Year)) %>%
  mutate(total_N_toenv2010_2014 = years_active_2010_2014 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2010_2014 = years_active_2010_2014 * 365 * Effluent) %>%
  mutate(Nflux_kgpd_5yr_avg = total_N_toenv2010_2014 / (365 *5)) %>%
  mutate(Effluent_galpd_5yr_avg = total_Eff_toenv2010_2014 / (365*5)) %>%
  st_transform(., crs = crs(ocean_mask_oahu))# this accounts for OSDS that were active for only part of

```

```

max(osds_2010_2014$Effluent_galpday_5yr_avg, na.rm = T) #14150
min(osds_2010_2014$Effluent_galpday_5yr_avg, na.rm = T) #60
mean(osds_2010_2014$Effluent_galpday_5yr_avg, na.rm = T) # 499
median(osds_2010_2014$Effluent_galpday_5yr_avg, na.rm = T) #420

osds_2010_2014_IW <- oahu_2023_cl %>%
  filter(Year < 2015 | is.na(Year)) %>% # this grabs the iws
  mutate(Year = ifelse(is.na(Year),2009,Year)) %>% # this just allows us to account IW for all POI
  #filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2010_2014 = (2015 - Year) - (2010 - Year)) %>%
  mutate(total_N_toenv2010_2014 = years_active_2010_2014 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2010_2014 = years_active_2010_2014 * 365 * Effluent) %>%
  mutate(Nflux_kgpday_5yr_avg = total_N_toenv2010_2014 / (365 * 5)) %>%
  mutate(Effluent_galpday_5yr_avg = total_Eff_toenv2010_2014 / (365 * 5)) %>%
  st_transform(., crs = crs(ocean_mask_oahu))

max(osds_2010_2014_IW$Effluent_galpday_5yr_avg, na.rm = T) #548841.2
min(osds_2010_2014_IW$Effluent_galpday_5yr_avg, na.rm = T) #60
mean(osds_2010_2014_IW$Effluent_galpday_5yr_avg, na.rm = T) #2716.339
median(osds_2010_2014_IW$Effluent_galpday_5yr_avg, na.rm = T) #420

# copying some cool code just for future reference

# library(magrittr)
#
# osds_2010_2014_IW <- oahu_2023_cl %>%
#   subset(Year < 2015 | is.na(Year)) %>%
#   within({
#     Year <- as.numeric(Year)
#     years_active_2010_2014 <- (2015 - Year) - (2010 - Year)
#     total_N_toenv2010_2014 <- years_active_2010_2014 * 365 * Nitrogen_Flux
#     total_Eff_toenv2010_2014 <- years_active_2010_2014 * 365 * Effluent
#     Nflux_kgpday_5yr_avg <- total_N_toenv2010_2014 / (365 * 5)
#     Effluent_galpday_5yr_avg <- total_Eff_toenv2010_2014 / (365 * 5)
#   })

```

Maui Nui Dataset

```

maui_gdb <- here("data/osds/Final_MauiNui_OSDS_Products.gdb")
st_layers(maui_gdb)
mauinui_2023 <- st_read(osds_gdb, layer = "MauiNui_OSDS_2023")

plot(mauiui_2023)

# check valid geoms

mauinui_2023 <- mauinui_2023 %>%

```

```

mutate(valid = st_is_valid(.),
       empty = st_is_empty(.))

maui_empty <- mauinui_2023 %>%
  filter(empty == TRUE) # none are empty, nice

# check stats

min(mauinui_2023$Effluent, na.rm = T) # 0 , and check NA
max(mauinui_2023$Effluent, na.rm = T) # 4393710
mean(mauinui_2023$Effluent, na.rm = T) #1192.007
median(mauinui_2023$Effluent, na.rm = T) #315

# nas?

mauinui_nas <- mauinui_2023 %>%
  filter(is.na(Effluent)) # 0

mauinui_zeros <- mauinui_2023 %>%
  filter(Effluent == 0) # 2 observations

# more checks

length(mauinui_2023$Effluent)
sum(is.infinite(mauinui_2023$Effluent))
sum(is.nan(mauinui_2023$Effluent))

# summarize all potential weirdness

mauinui_df <- st_drop_geometry(mauinui_2023)

summary_check <- function(df) {
  data.frame(
    Column = names(df),
    NA_count = sapply(df, function(x) sum(is.na(x))),
    Inf_count = sapply(df, function(x) sum(is.infinite(x))),
    NaN_count = sapply(df, function(x) sum(is.nan(x)))
  )
}

mauinui_check <- summary_check(mauinui_df)

# all of the NAs in effluent are new. what percentage of them are NA?

mauinui_new_osds <- mauinui_2023 %>%
  filter(comments == "new OSDS for 2023 update") # 638

mapview(mauinui_nas, col.regions = "green")+
  mapview(mauinui_zeros, col.regions = "blue")

# how many injection wells?

```

```

mauinui_iw <- mauinui_2023 %>%
  filter(PermitType == "Injection Well")

mapview(mauinui_iw)

# clean version

maui_2023_cl <- mauinui_2023 %>%
  filter(!is.na(Effluent))

# subset points by injection wells versus no injection wells for each 5 year interval

mauinui_2010_2014 <- maui_2023_cl %>%
  filter(Year < 2015) %>%
  filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2010_2014 = (2015 - Year) - (2010 - Year)) %>%
  mutate(total_N_toenv2010_2014 = years_active_2010_2014 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2010_2014 = years_active_2010_2014 * 365 * Effluent) %>%
  mutate(Nflux_kgpd_5yr_avg = total_N_toenv2010_2014 / (365 * 5)) %>%
  mutate(Effluent_galpd_5yr_avg = total_Eff_toenv2010_2014 / (365 * 5)) %>%
  st_transform(., crs = crs(ocean_mask))

mean(mauinui_2010_2014$Effluent) #682.72
max(mauinui_2010_2014$Effluent) #27000

mauinui_2010_2014_IW <- maui_2023_cl %>%
  filter(Year < 2015) %>%
  mutate(Year = ifelse(is.na(Year), 2009, Year)) %>% # this just allows us to account IW for all POI
  #filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2010_2014 = (2015 - Year) - (2010 - Year)) %>%
  mutate(total_N_toenv2010_2014 = years_active_2010_2014 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2010_2014 = years_active_2010_2014 * 365 * Effluent) %>%
  mutate(Nflux_kgpd_5yr_avg = total_N_toenv2010_2014 / (365 * 5)) %>%
  mutate(Effluent_galpd_5yr_avg = total_Eff_toenv2010_2014 / (365 * 5)) %>%
  st_transform(., crs = crs(ocean_mask))

mean(mauinui_2010_2014_IW$Effluent) #727
max(mauinui_2010_2014_IW$Effluent) #27000

mauinui_2015_2019 <- maui_2023_cl %>%
  filter(Year < 2020) %>%
  filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2010_2014 = (2020 - Year) - (2015 - Year)) %>%
  mutate(total_N_toenv2010_2014 = years_active_2010_2014 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2010_2014 = years_active_2010_2014 * 365 * Effluent) %>%
  mutate(Nflux_kgpd_5yr_avg = total_N_toenv2010_2014 / (365 * 5)) %>%
  mutate(Effluent_galpd_5yr_avg = total_Eff_toenv2010_2014 / (365 * 5)) %>%
  st_transform(., crs = crs(ocean_mask))

```

```

mean(mauinui_2015_2019$Effluent) # 607
max(mauinui_2015_2019$Effluent)

mauinui_2015_2019_IW <- mau_2023_c1 %>%
  filter(Year < 2020) %>%
  mutate(Year = ifelse(is.na(Year),2009,Year)) %>% # this just allows us to account IW for all POI
  #filter(PermitType != "Injection Well") %>%
  mutate(Year = as.numeric(Year)) %>%
  mutate(years_active_2010_2014 = (2020 - Year) - (2015 - Year)) %>%
  mutate(total_N_toenv2010_2014 = years_active_2010_2014 * 365 * Nitrogen_Flux) %>%
  mutate(total_Eff_toenv2010_2014 = years_active_2010_2014 * 365 * Effluent) %>%
  mutate(Nflux_kgpd5yr_avg = total_N_toenv2010_2014 / (365 *5)) %>%
  mutate(Effluent_galpd5yr_avg = total_Eff_toenv2010_2014 / (365*5)) %>%
  st_transform(., crs = crs(ocean_mask))

mean(mauinui_2015_2019_IW$Effluent) #632

```

Step 1: Aggregate effluent and convert to 100 x 100 m raster

Goal: - Sum up effluent from points within each 100 x 100 grid cell to get total osds effluent for that pixel

Assumptions:

```

# read in ocean mask and use extent of oahu and maui points to crop
ocean_mask <- terra::rast(file.path(data_dir,"OceanMask_LBSP_post2018.tif")) #

osds_2015_2019_nad83 <- st_transform(osds_2015_2019, crs = crs(ocean_mask))

maui_osds_nad83 <- st_transform(mauinui_2023, crs = crs(ocean_mask))

# create oahu specific ocean mask and expand extent to fit entire ocean mask
extent <- ext(oahu_2023_nad83)
extent_vec <- as.vector(extent)

add_vec <- c(-14283,14300,-20892,20000)
new_extent <- extent_vec + add_vec
expanded <- ext(new_extent)

ocean_mask_oahu <- terra::crop(ocean_mask, expanded, mask = T)
plot(ocean_mask_oahu)

# repeat for maui

mauinui_ext <- ext(maui_osds_nad83)
extent_vec <- as.vector(mauinui_ext)

add_vec <- c(-20000,20000,-30000,20000)
new_maui_ext <- extent_vec + add_vec
maui_expanded <- ext(new_maui_ext)

ocean_mask_maui <- terra::crop(ocean_mask, maui_expanded, mask = T)
plot(ocean_mask_maui)

```

```

# rasterize function, to apply to multiple point subsets

rasterizePoints <- function(points,mask){

  # create raster template and rasterize points with terra
  r_om <- rast(mask)
  #rasterize points, using sum to aggregate effluent
  osds_raster <- rasterize(vect(points), r_om, field = "Effluent_galpday_5yr_avg", fun = "sum", na.rm =

  return(osds_raster)

}

# apply to points for different scenarios

oahu_2015_2019_eff <- rasterizePoints(osds_2015_2019_nad83,ocean_mask_oahu)
oahu_2015_2019_IWeff <- rasterizePoints(osds_2015_2019_IW,ocean_mask_oahu)
oahu_2010_2014_eff <- rasterizePoints(osds_2010_2014,ocean_mask_oahu)
oahu_2010_2014_IWeff <- rasterizePoints(osds_2010_2014_IW,ocean_mask_oahu)

plot(oahu_2015_2019_eff)
plot(oahu_2010_2014_eff)
plot(oahu_2010_2014_IWeff)

# repeat with maui

maui_2010_2014_eff <- rasterizePoints(mauinui_2010_2014,ocean_mask_maui)
maui_2010_2014_IWeff <- rasterizePoints(mauinui_2010_2014_IW,ocean_mask_maui)
maui_2015_2019_eff <- rasterizePoints(mauinui_2015_2019, ocean_mask_maui)
maui_2015_2019_IWeff <- rasterizePoints(mauinui_2015_2019_IW, ocean_mask_maui)

plot(maui_2010_2014_eff)
plot(maui_2010_2014_IWeff)
plot(maui_2015_2019_eff)
plot(maui_2015_2019_IWeff)

minmax(maui_2010_2014_eff)

```

Step 2: Apply focal statistics to estimate “sphere of impact” into nearshore

Goal:

- Estimate how OSDS impacts the nearshore

Assumptions:

- Ocean areas closest to pixels with higher effluent, from aggregated OSDS point data, will experience the highest impact from the leaching / runoff

```

# prep and funciton for Oahu

```



```

ocean_mask_oahu[ocean_mask_oahu != 10] <- NA
ocean_mask_maui[ocean_mask_maui != 10] <- NA

applyFocalStats_extractByMask <- function(raster,mask){

  # create a matrix of the raster values, and then replace any non zero value with 1
  fw_mat <- focalMat(raster,2500, type = "circle") #2100 does not cover all indicator maps
  fw_mat[fw_mat > 0 ] <- 1

# apply focal statistics
  focalStat <- terra::focal(raster, w = fw_mat, fun = "sum", na.rm = T) # na.rm key to making it expand

# Extract by ocean mask
  focalStat_nearshore <- mask(focalStat, mask)

  return(focalStat_nearshore)
}

# apply to scenario

focStat_2015_2019 <- applyFocalStats_extractByMask(oahu_2015_2019_eff,ocean_mask_oahu)
plot(focStat_2015_2019)
focStat_2015_2019_IW <- applyFocalStats_extractByMask(oahu_2015_2019_IWeff,ocean_mask_oahu)
plot(focStat_2015_2019_IW)
focStat_2010_2014 <- applyFocalStats_extractByMask(oahu_2010_2014_eff,ocean_mask_oahu)
plot(focStat_2010_2014)
focStat_2010_2014_IW <- applyFocalStats_extractByMask(oahu_2010_2014_IWeff,ocean_mask_oahu)
plot(focStat_2010_2014_IW)

# maui

maui_focStat_2010_2014 <- applyFocalStats_extractByMask(maui_2010_2014_eff, ocean_mask_maui)
plot(maui_focStat_2010_2014)
maui_focStat_2010_2014IW <- applyFocalStats_extractByMask(maui_2010_2014_IWeff, ocean_mask_maui)
plot(maui_focStat_2010_2014IW)
maui_focStat_2015_2019 <- applyFocalStats_extractByMask(maui_2015_2019_eff, ocean_mask_maui)
plot(maui_focStat_2015_2019)
maui_focStat_2015_2019IW <- applyFocalStats_extractByMask(maui_2015_2019_IWeff, ocean_mask_maui)
plot(maui_focStat_2015_2019IW)

minmax(maui_focStat_2010_2014)
# 101060??

minmax(maui_focStat_2015_2019)

```

Step 3: Use ocean mask to limit area of impact to marine space & add decay function

Assumptions:

- Intensity of impact lessens with distance from shore, due to chemical decay of N or P and / or wave or currents
- Decay occurs as a linear function of distance from shore

```
# get custom decay function

source(here("workflow/offshoreDecay.R")) # identical to all other decay for LBSP

# read in Euclidean distance, check the crs and crop by the custom extent

euc_distance <- rast(file.path(data_dir,"DistFromShore_Euc_2023.tif"))
euc_distance_maui <- terra::crop(euc_distance, ocean_mask_maui, mask = T)
plot(euc_distance_maui)

# apply to nearshore rasters - Oahu

final_osds_2015_2019 <- offshore_decay(focStat_2015_2019,euc_distance_oahu)
final_osds_2015_2019[final_osds_2015_2019 == 0] <- NA
plot(final_osds_2015_2019)

final_osds_2015_2019_IW <- offshore_decay(focStat_2015_2019_IW,euc_distance_oahu)
final_osds_2015_2019_IW[final_osds_2015_2019_IW == 0] <- NA
plot(final_osds_2015_2019_IW)

final_osds_2010_2014 <- offshore_decay(focStat_2010_2014,euc_distance_oahu)
final_osds_2010_2014[final_osds_2010_2014 == 0] <- NA
plot(final_osds_2010_2014)

final_osds_2010_2014_IW <- offshore_decay(focStat_2010_2014_IW,euc_distance_oahu)
final_osds_2010_2014_IW[final_osds_2010_2014_IW == 0] <- NA
plot(final_osds_2010_2014_IW)

# apply to nearshore rasters - Maui

maui_final_osds_2010_2014 <- offshore_decay(maui_focStat_2010_2014,euc_distance_maui)
maui_final_osds_2010_2014[maui_final_osds_2010_2014 == 0] <- NA
plot(maui_final_osds_2010_2014)

maui_final_osds_2010_2014_IW <- offshore_decay(maui_focStat_2010_2014IW,euc_distance_maui)
maui_final_osds_2010_2014_IW[maui_final_osds_2010_2014_IW == 0] <- NA
plot(maui_final_osds_2010_2014_IW)

maui_final_osds_2015_2019 <- offshore_decay(maui_focStat_2015_2019,euc_distance_maui)
maui_final_osds_2015_2019[maui_final_osds_2015_2019 == 0] <- NA
plot(maui_final_osds_2015_2019)

maui_final_osds_2015_2019_IW <- offshore_decay(maui_focStat_2015_2019IW,euc_distance_maui)
maui_final_osds_2015_2019_IW[maui_final_osds_2015_2019_IW == 0] <- NA
plot(maui_final_osds_2015_2019_IW)
```

```
minmax(maui_final_osds_2010_2014)
minmax(maui_final_osds_2015_2019)
```

Final Step: Save Rasters

```
writeRaster(
  final_osds_2015_2019,
  filename = here("outputs/Oahu OSDS_2015_2019_eff.tif"),
  overwrite = T
)

writeRaster(
  final_osds_2015_2019_IW,
  filename = here("outputs/Oahu OSDS_2015_2019_IWeff.tif"),
  overwrite = T
)

writeRaster(
  final_osds_2010_2014,
  filename = here("outputs/Oahu OSDS_2010_2014_eff.tif"),
  overwrite = T
)

writeRaster(
  final_osds_2010_2014_IW,
  filename = here("outputs/Oahu OSDS_2010_2014_IWeff.tif"),
  overwrite = T
)

# maui

writeRaster(
  maui_final_osds_2010_2014,
  filename = here("outputs/Maui OSDS_2010_2014_eff.tif"),
  overwrite = T
)

writeRaster(
  maui_final_osds_2010_2014_IW,
  filename = here("outputs/Maui OSDS_2010_2014_IWeff.tif"),
  overwrite = T
)

writeRaster(
  maui_final_osds_2015_2019,
  filename = here("outputs/Maui OSDS_2015_2019_eff.tif"),
  overwrite = T
)
```

```
writeRaster(  
  maui_final_osds_2015_2019_IW,  
  filename = here("outputs/Maui_OSDS_2015_2019_IWeff.tif"),  
  overwrite = T  
)
```